Biochemistry Cumulative Examination

October 25, 1997

Enzymes can carry out an amazing variety of chemistry just with the functional groups of the twenty amino acids and the occasional metal ion. However, there are times when more complicated functional groups are required to do certain kinds of reactions and coenzymes (or cofactors, if you prefer) are required. There are a very large number of coenzymes and knowledge of their function in biochemistry is important.

1a. Name six different coenzymes. Name them as fully as possible (*i.e.* cryptic abbreviations, acronyms, or designations will not carry full credit). (5 points each)

1b. For each coenzyme you name describe what kind of reaction or functional group transformation it helps enzymes carry out. (5 points each)

2. For any two coenzymes draw the structure of the "business end" of the molecule. In other words, you need not show parts of the molecule important for binding but that take no part in the chemistry of the coenzyme and enzyme. (15 points each)

3. For any one coenzyme show a simple reaction mechanism. (10 points)